# Single Factor Test Report

Factor: alpha004

Rebalance Periods: 1, 5, 10 days

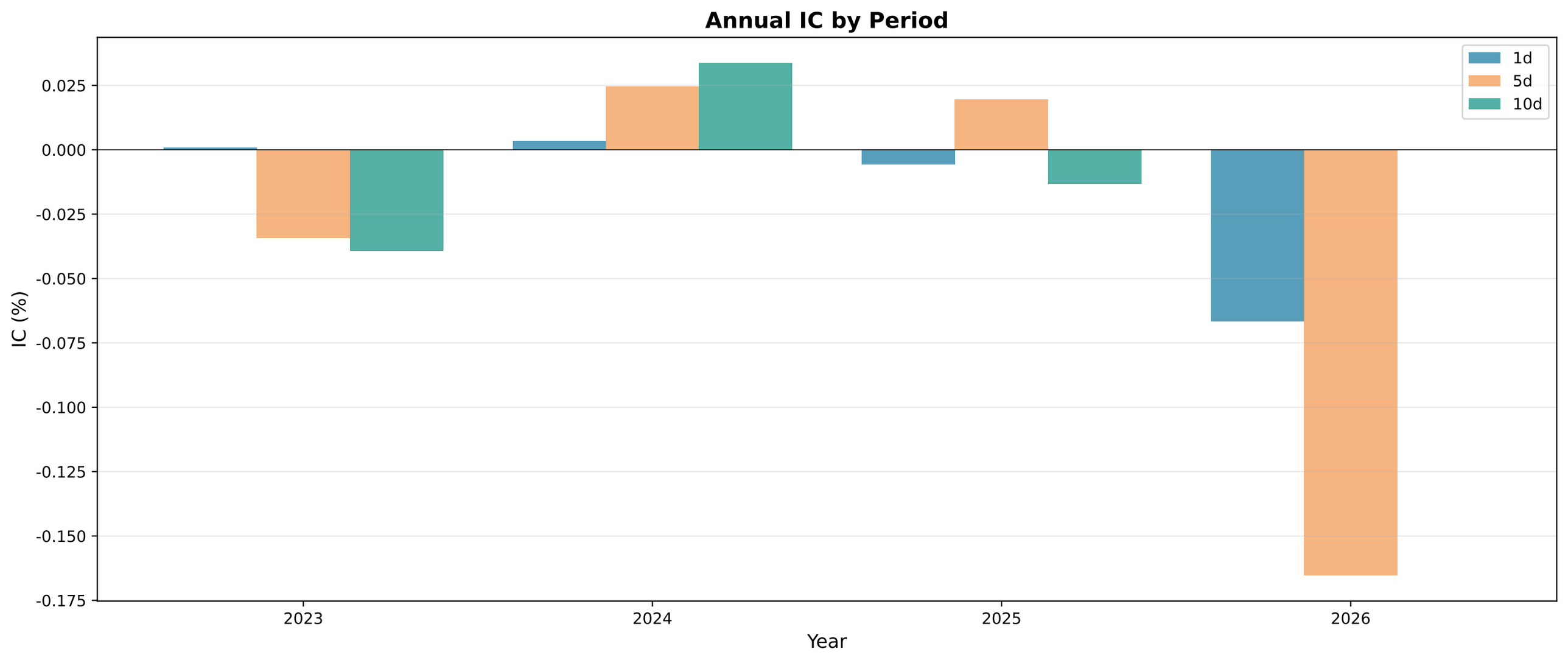
Generated: 2026-02-27 01:41:51

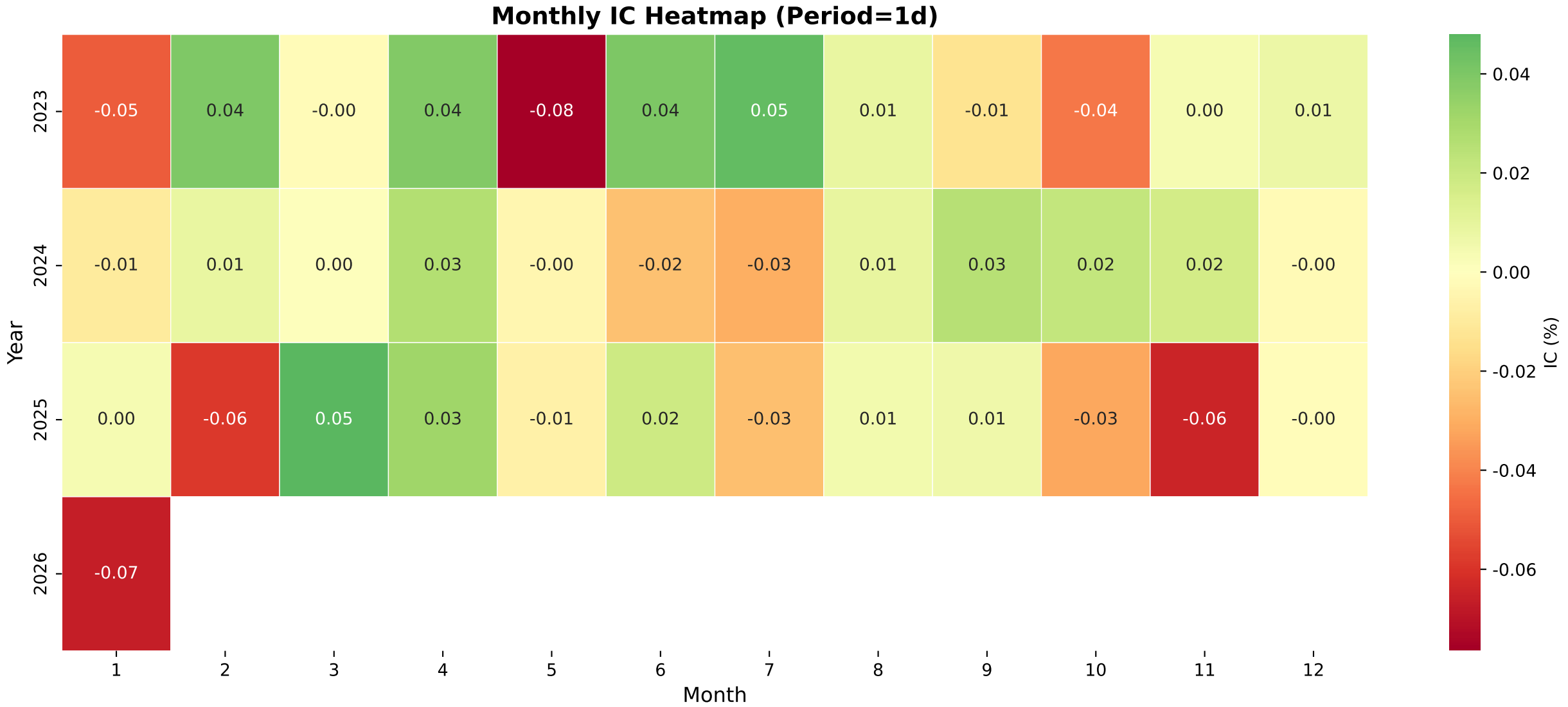
## Report Contents:

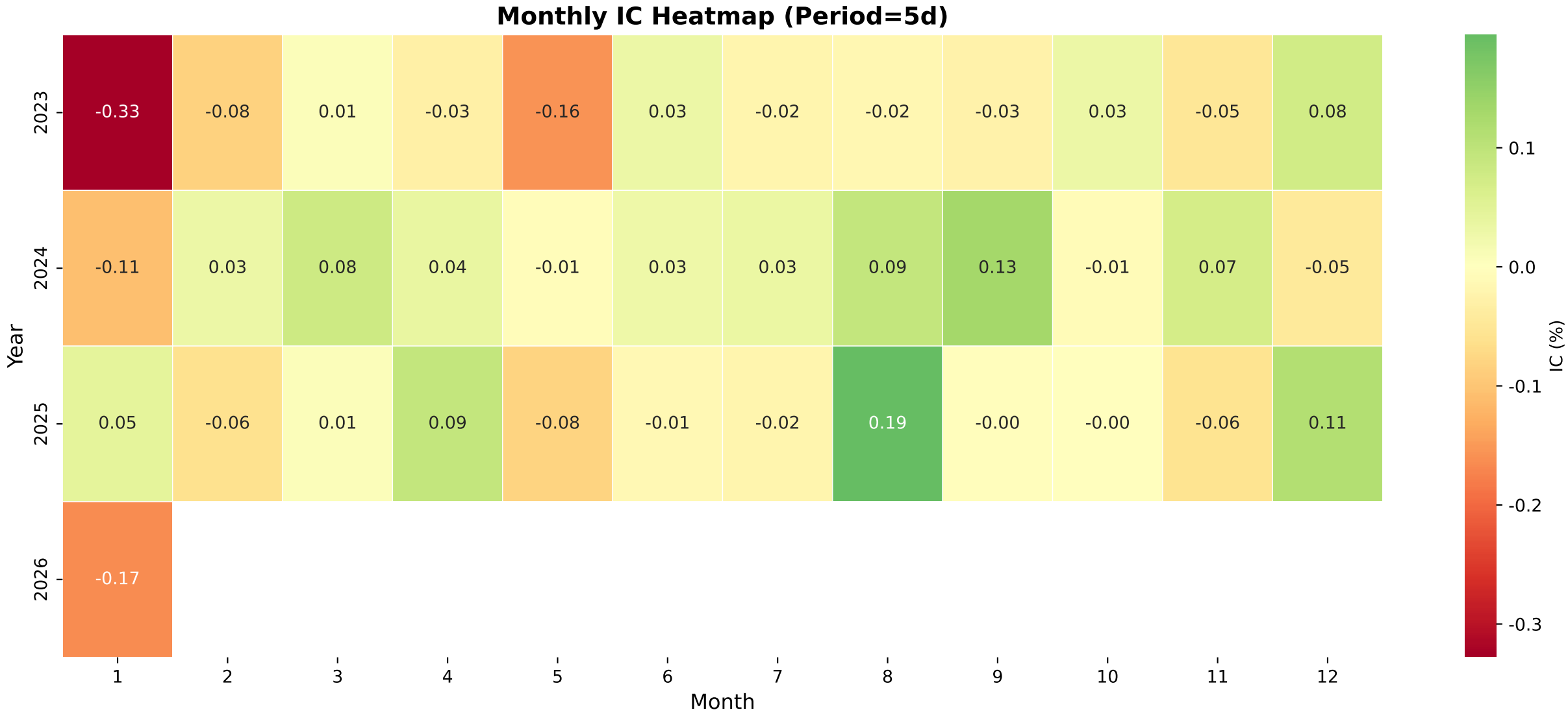
1. IC Analysis
2. Long-Short Backtest
3. Long-Only Backtest
4. Short-Only Backtest

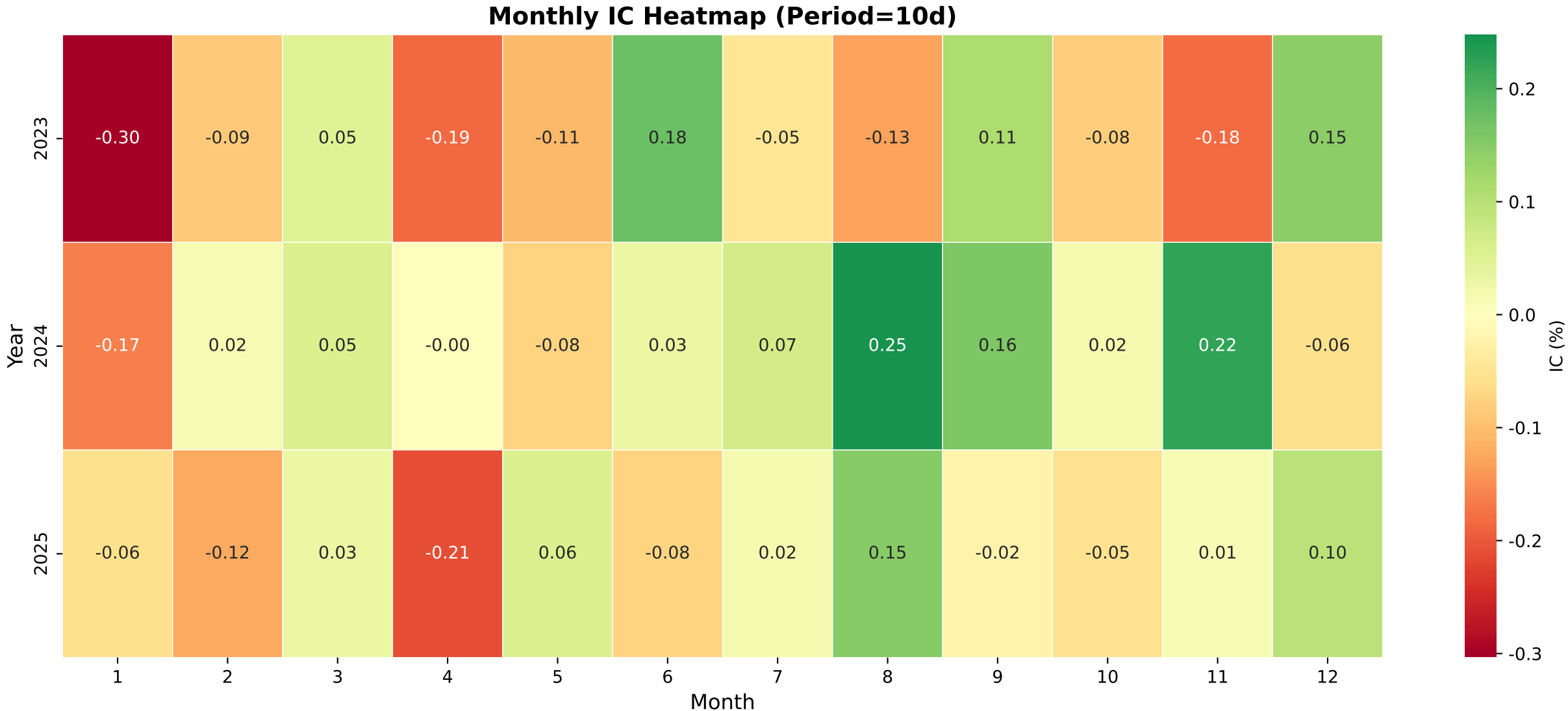
# IC Statistics

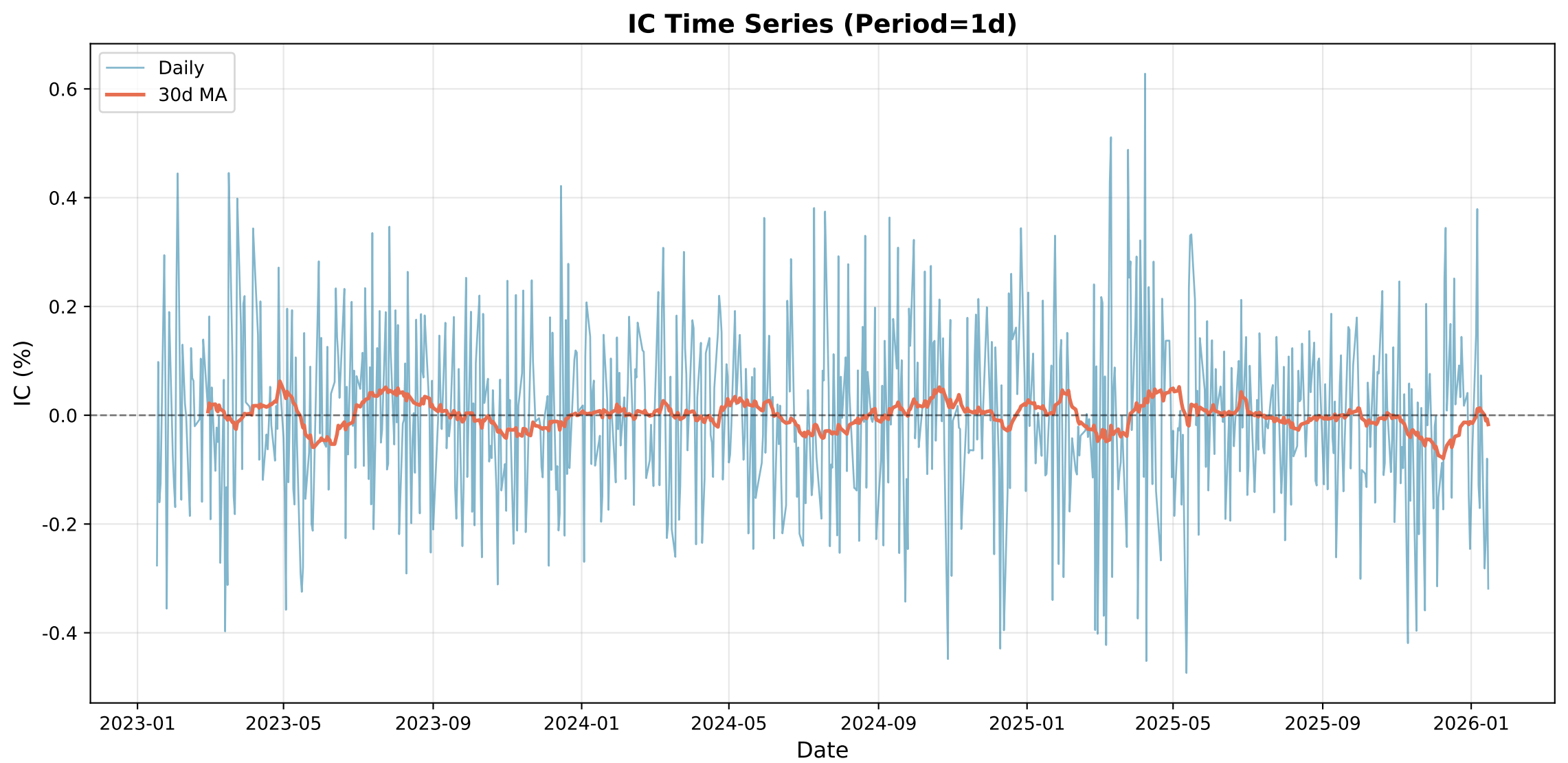
| Metric | Period_1 | Period_5 | Period_10 |
|---|---|---|---|
| Mean | -0.0015 | 0.0028 | -0.0052 |
| Std | 0.1642 | 0.1683 | 0.1699 |
| IR | -0.0089 | 0.0167 | -0.0307 |
| Skew | 0.0961 | -0.0483 | 0.2210 |
| Kurtosis | 0.2500 | -0.2925 | -0.1924 |
| t_value | -0.2443 | 0.2043 | -0.2657 |
| p_value | 0.807067 | 0.838376 | 0.791204 |
| Win_Rate | 0.4821 | 0.5133 | 0.4267 |
| IC>0.02 | 0.4369 | 0.4600 | 0.3867 |

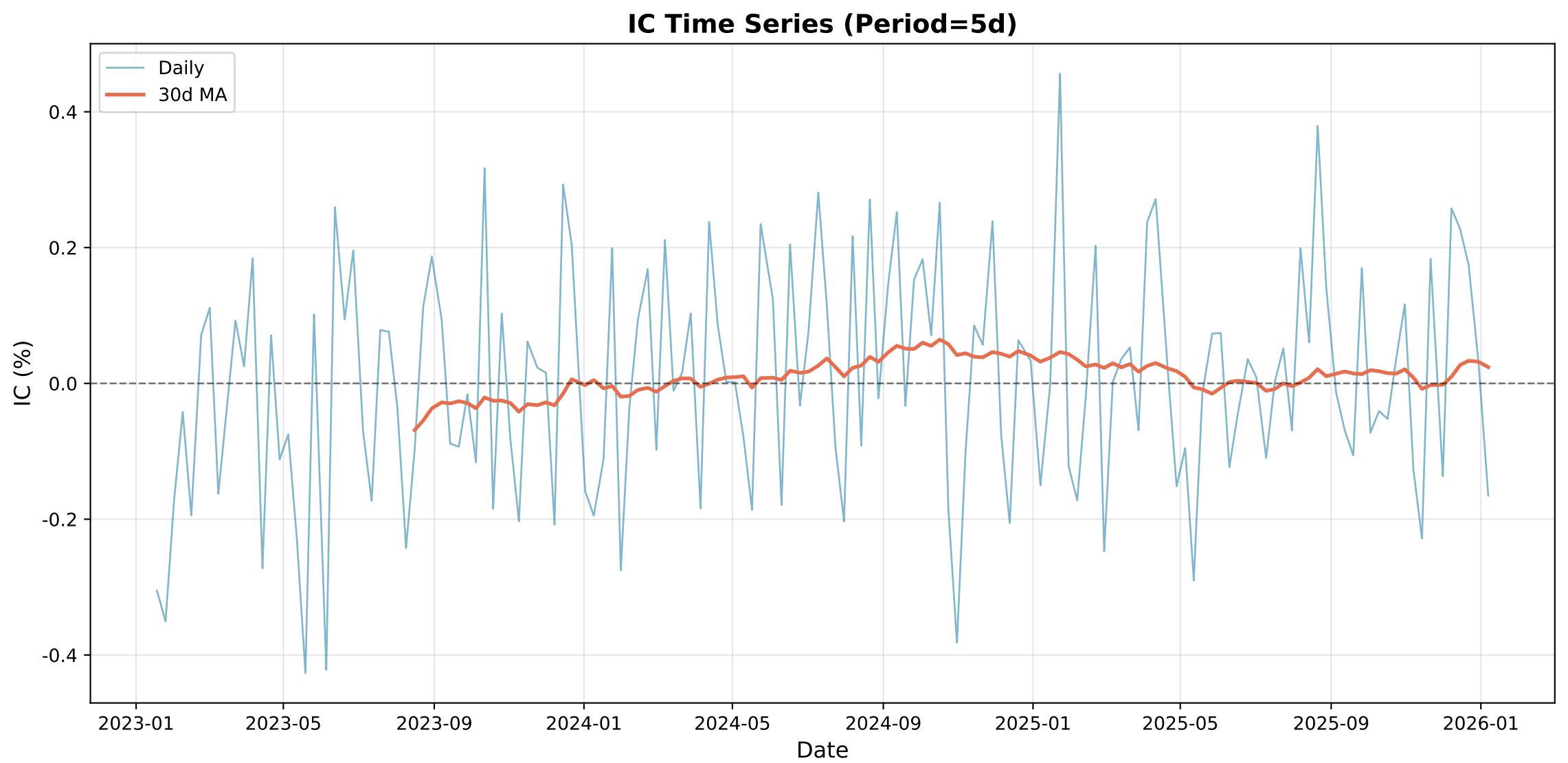

Annual IC by Period
1d
5d
10d
0.025
0.000
-0.025
-0.050
-0.075
-0.100
-0.125
-0.150
-0.175
IC (%)
2023
2024
2025
2026
Year

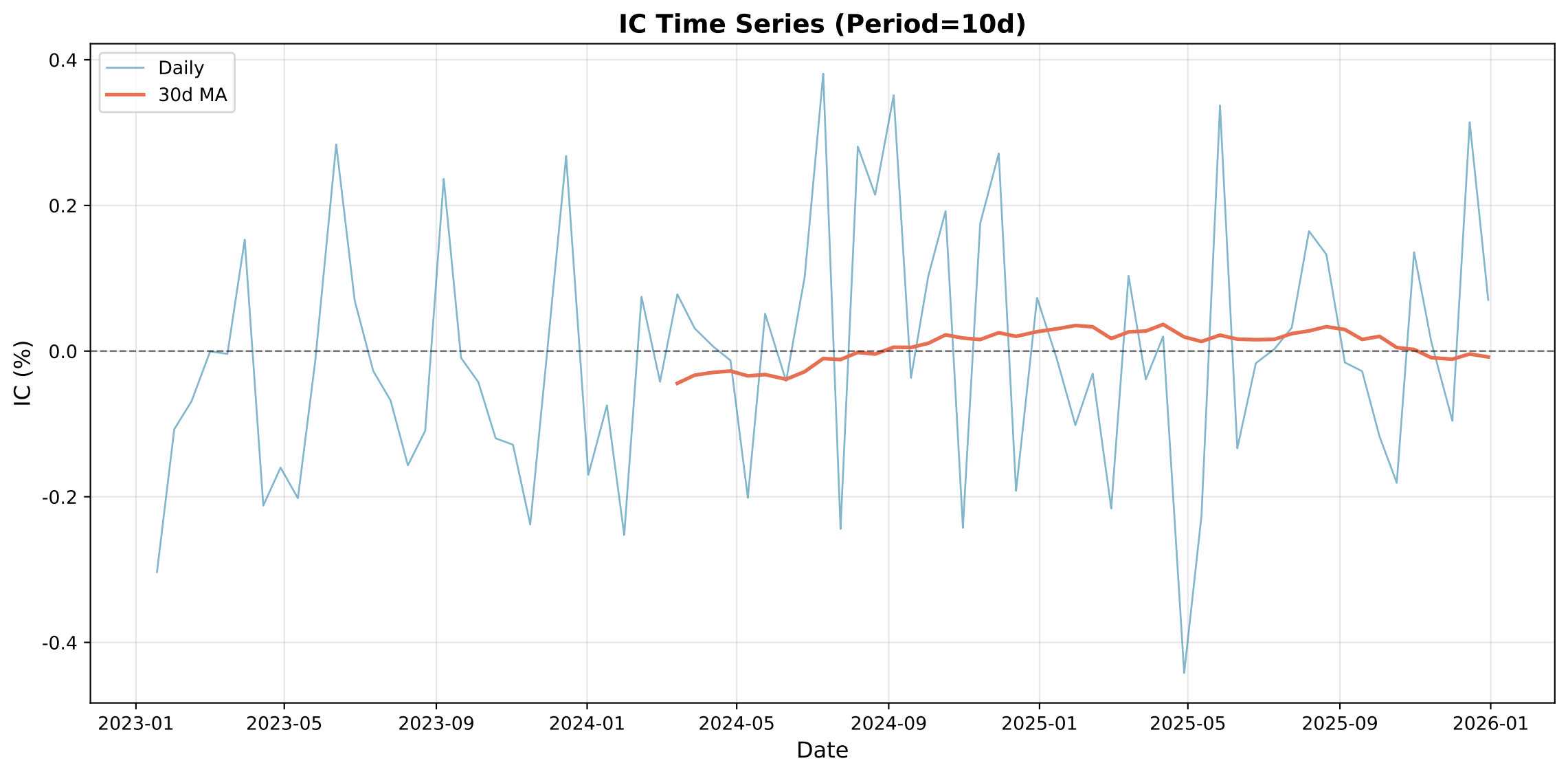

Monthly IC Heatmap (Period=1d)
Year
2023
2024
2025
2026
-0.05 0.04 -0.00 0.04 -0.08 0.04 0.05 0.01 -0.01 -0.04 0.00 0.01
-0.01 0.01 0.00 0.03 -0.00 -0.02 -0.03 0.01 0.03 0.02 0.02 -0.00
0.00 -0.06 0.05 0.03 -0.01 0.02 -0.03 0.01 0.01 -0.03 -0.06 -0.00
-0.07
1 2 3 4 5 6 7 8 9 10 11 12
Month
0.04
0.02
0.00
-0.02
-0.04
-0.06
IC (%)

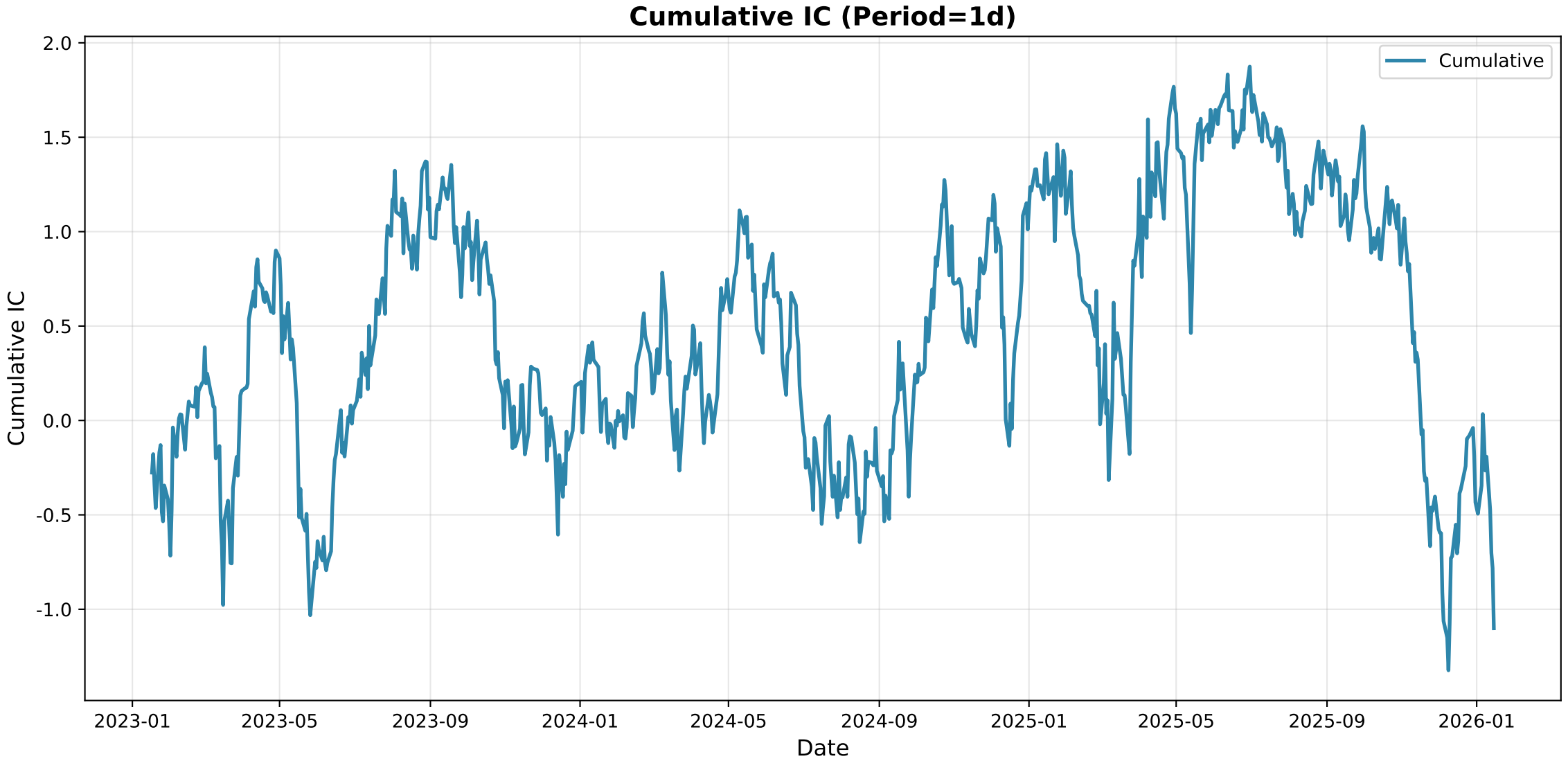

Monthly IC Heatmap (Period=5d)
Year
2023
2024
2025
2026
-0.33 -0.08 0.01 -0.03 -0.16 0.03 -0.02 -0.02 -0.03 0.03 -0.05 0.08
-0.11 0.03 0.08 0.04 -0.01 0.03 0.03 0.09 0.13 -0.01 0.07 -0.05
0.05 -0.06 0.01 0.09 -0.08 -0.01 -0.02 0.19 -0.00 -0.00 -0.06 0.11
-0.17
1 2 3 4 5 6 7 8 9 10 11 12
Month
IC (%)
0.1
0.0
-0.1
-0.2
-0.3

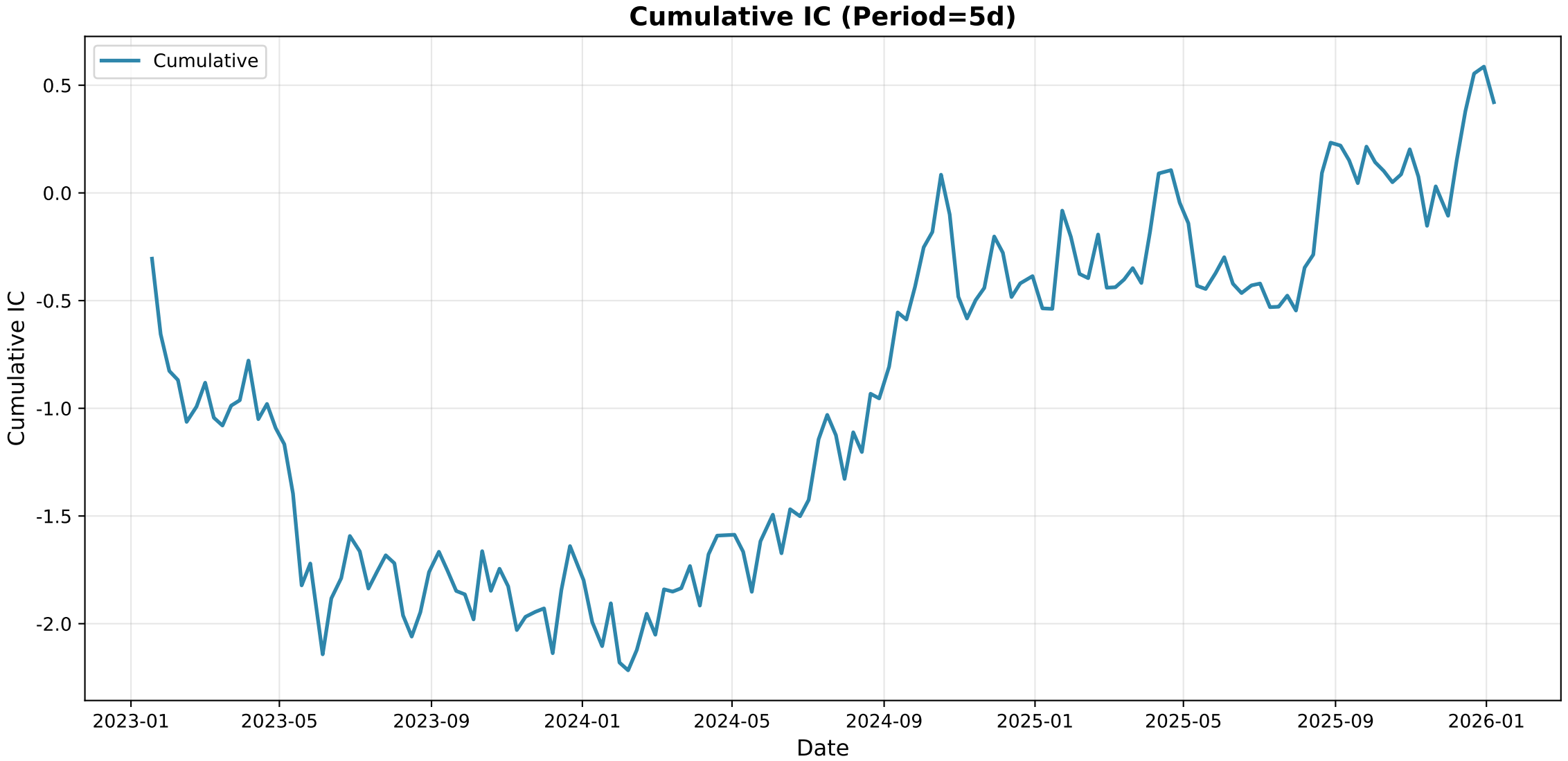

Monthly IC Heatmap (Period=10d)
Year
2023
2024
2025
-0.30 -0.09 0.05 -0.19 -0.11 0.18 -0.05 -0.13 0.11 -0.08 -0.18 0.15
-0.17 0.02 0.05 -0.00 -0.08 0.03 0.07 0.25 0.16 0.02 0.22 -0.06
-0.06 -0.12 0.03 -0.21 0.06 -0.08 0.02 0.15 -0.02 -0.05 0.01 0.10
1 2 3 4 5 6 7 8 9 10 11 12
Month
IC (%)
0.2
0.1
0.0
-0.1
-0.2
-0.3

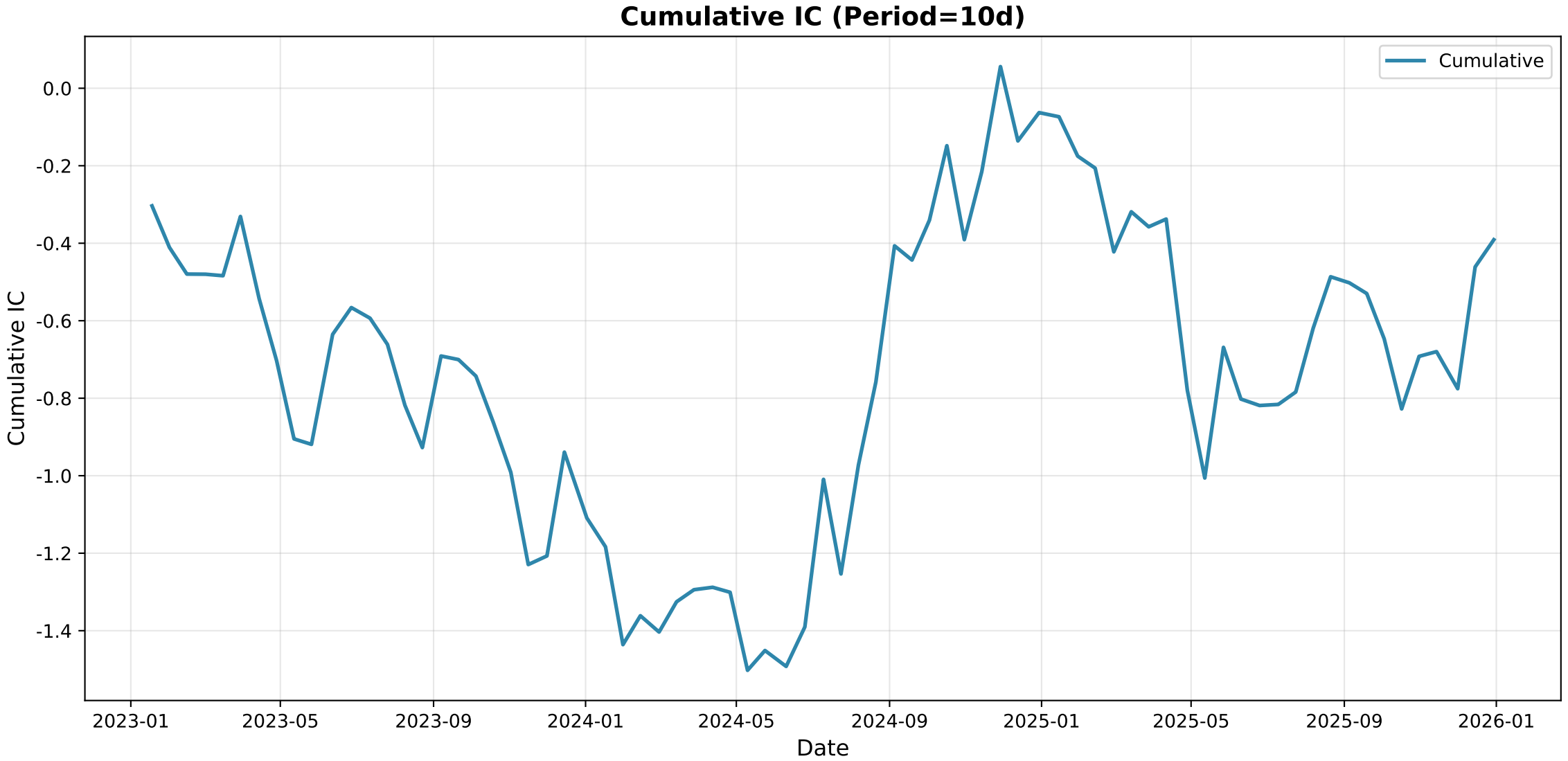

IC Time Series (Period=1d)
Daily
30d MA
IC (%)
0.6
0.4
0.2
0.0
-0.2
-0.4
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
Date


IC Time Series (Period=5d)
Daily
30d MA
0.4
0.2
0.0
-0.2
-0.4
IC (%)
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
Date


IC Time Series (Period=10d)
Daily
30d MA
IC (%)
0.4
0.2
0.0
-0.2
-0.4
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
Date

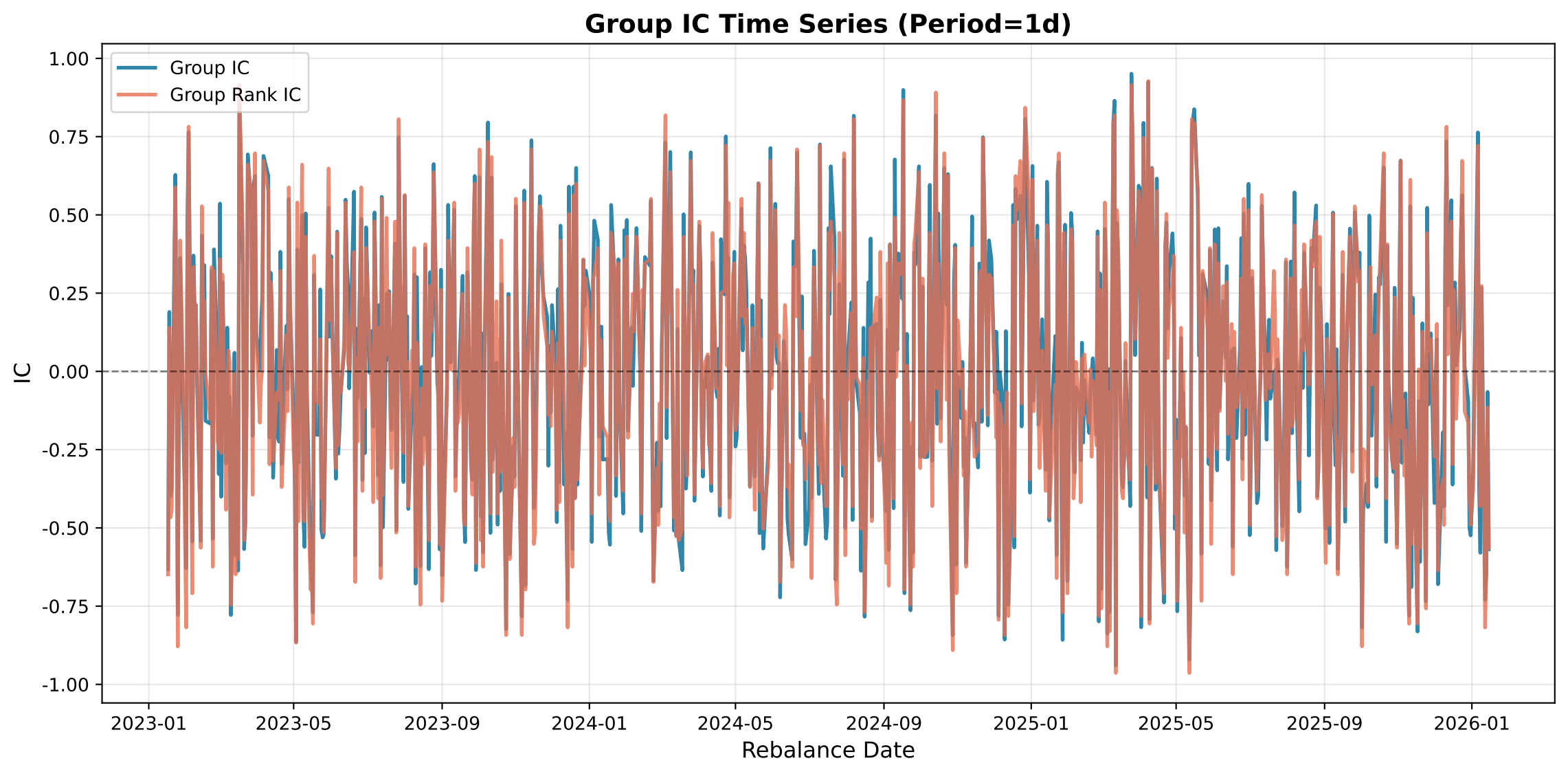

Cumulative IC (Period=1d)
Cumulative
Cumulative IC
2.0
1.5
1.0
0.5
0.0
-0.5
-1.0
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
Date

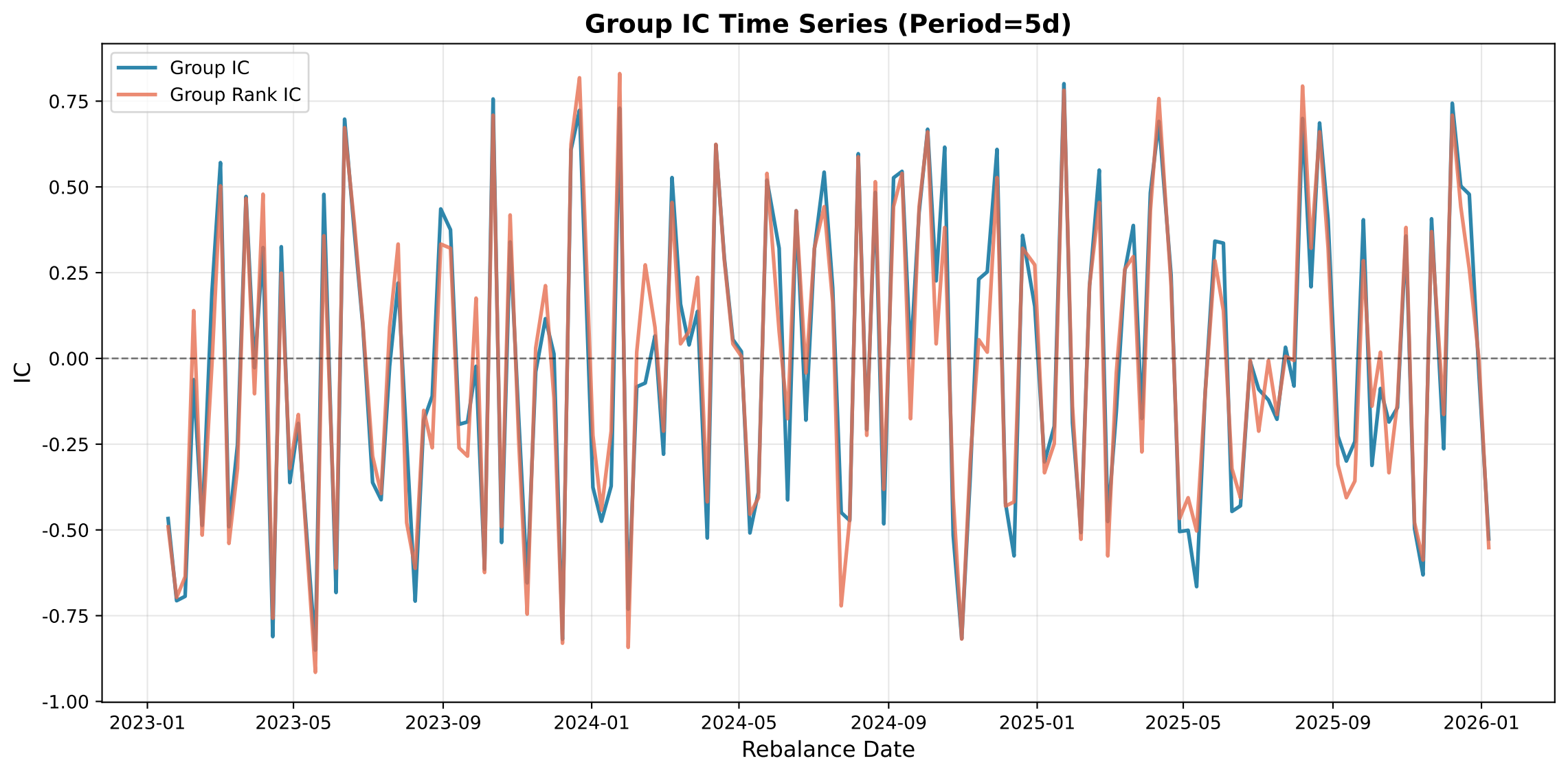

Cumulative IC (Period=5d)
Cumulative
Cumulative IC
0.5
0.0
-0.5
-1.0
-1.5
-2.0
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
Date

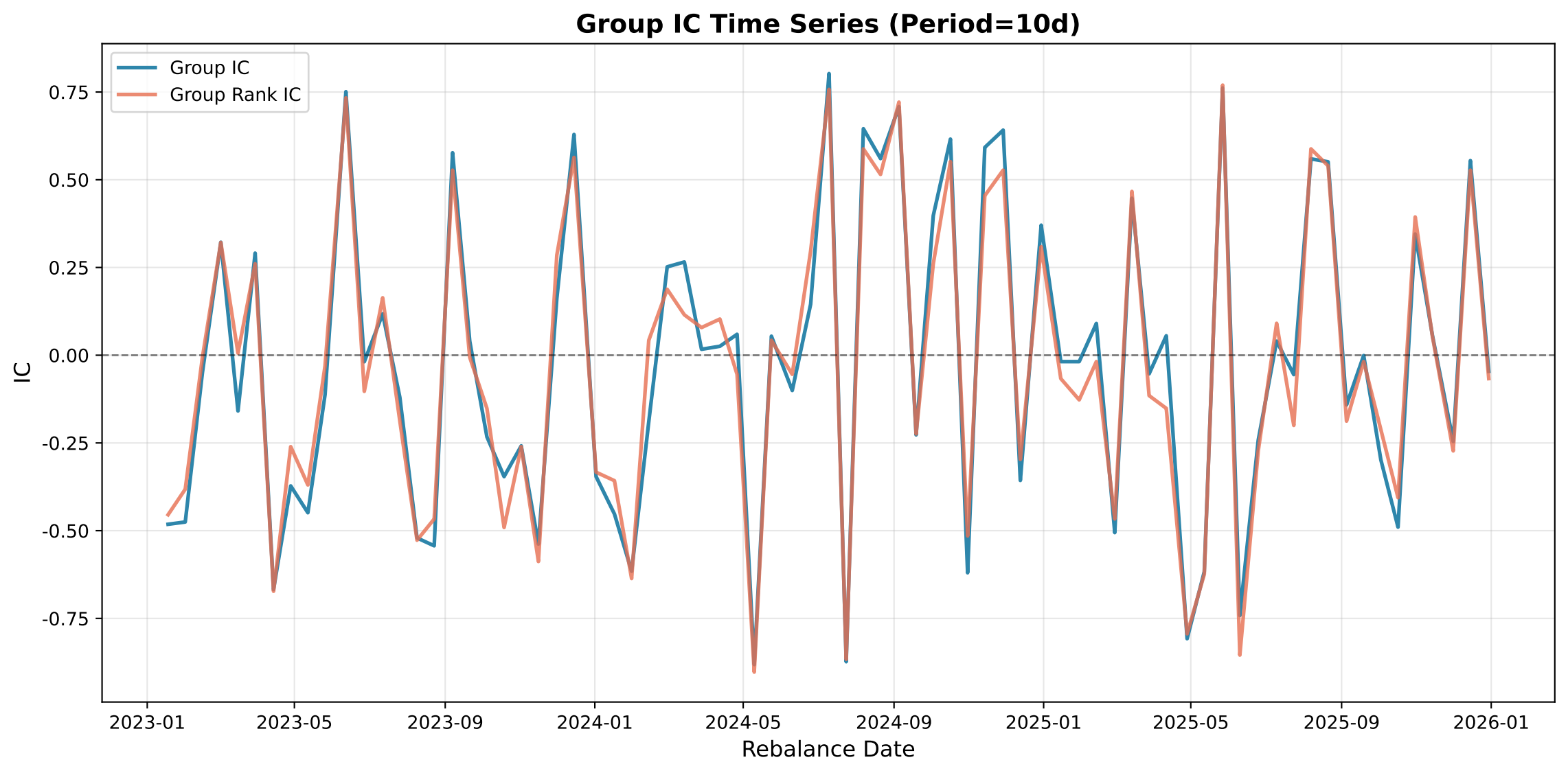

Cumulative IC (Period=10d)
Cumulative
Cumulative IC
0.0
-0.2
-0.4
-0.6
-0.8
-1.0
-1.2
-1.4
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
Date

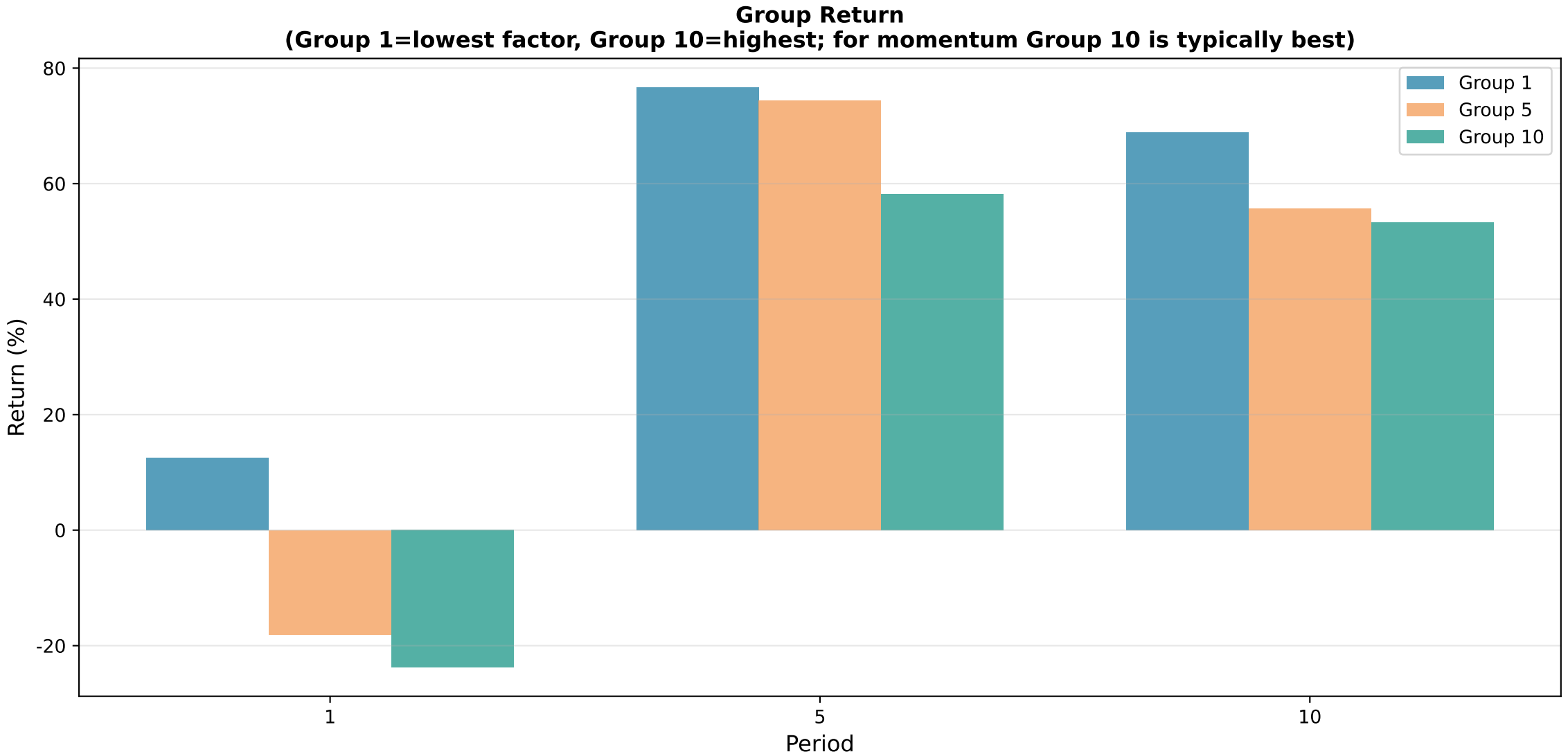
# Group IC Statistics

| Metric | Period_1 | Period_5 | Period_10 |
|---|---|---|---|
| Mean | -0.0095 | -0.0142 | -0.0238 |
| Std | 0.4139 | 0.4351 | 0.4405 |
| IR | -0.0228 | -0.0327 | -0.0540 |
| t_value | -0.6267 | -0.3999 | -0.4680 |
| p_value | 0.531068 | 0.689814 | 0.641188 |
| Win_Rate | 0.4927 | 0.4600 | 0.4533 |


Group IC Time Series (Period=1d)
Group IC
Group Rank IC
IC
1.00
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
Rebalance Date

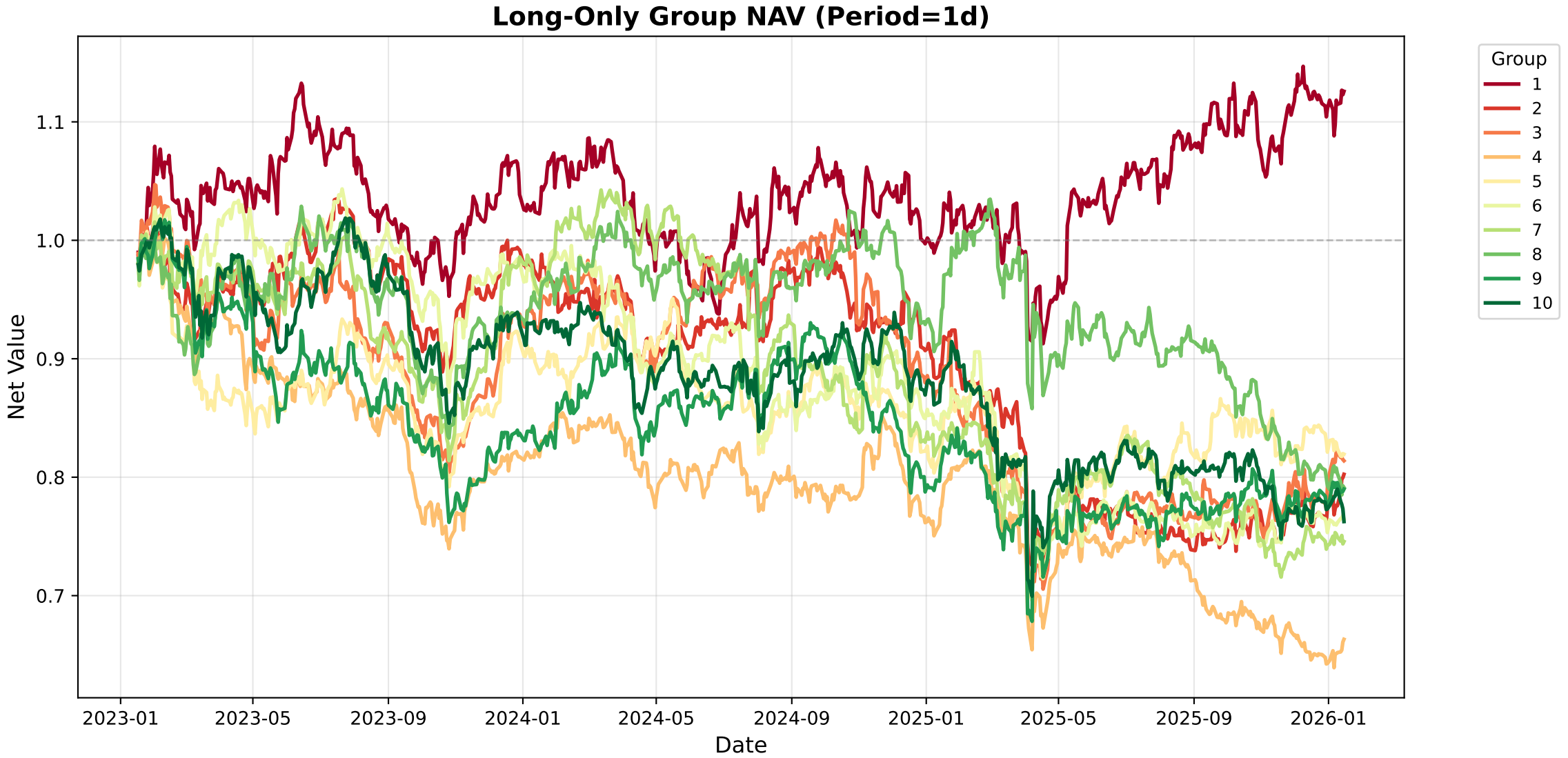

Group IC Time Series (Period=5d)
Group IC
Group Rank IC
IC
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
Rebalance Date

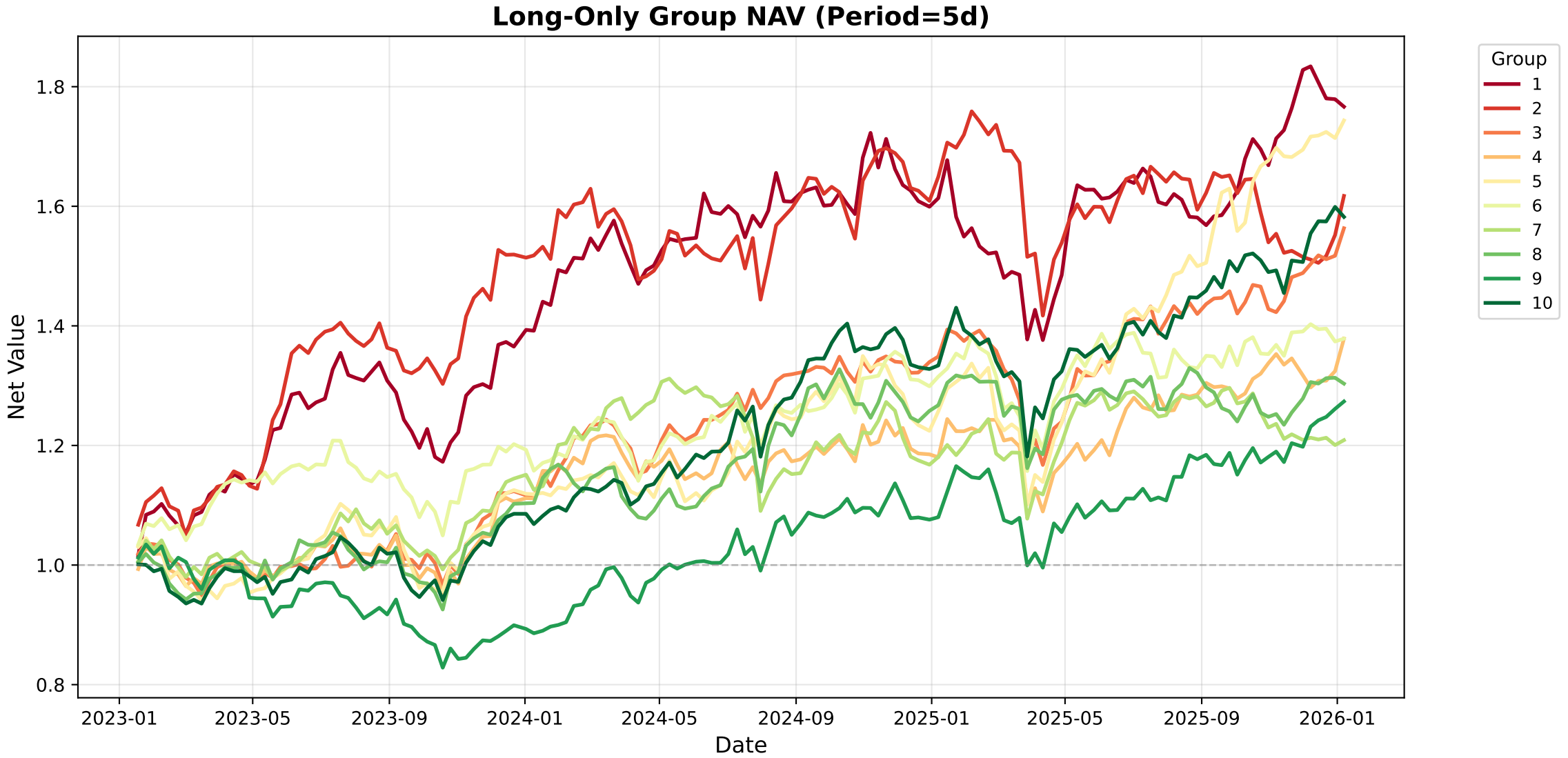

Group IC Time Series (Period=10d)
Group IC
Group Rank IC
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
IC
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
Rebalance Date

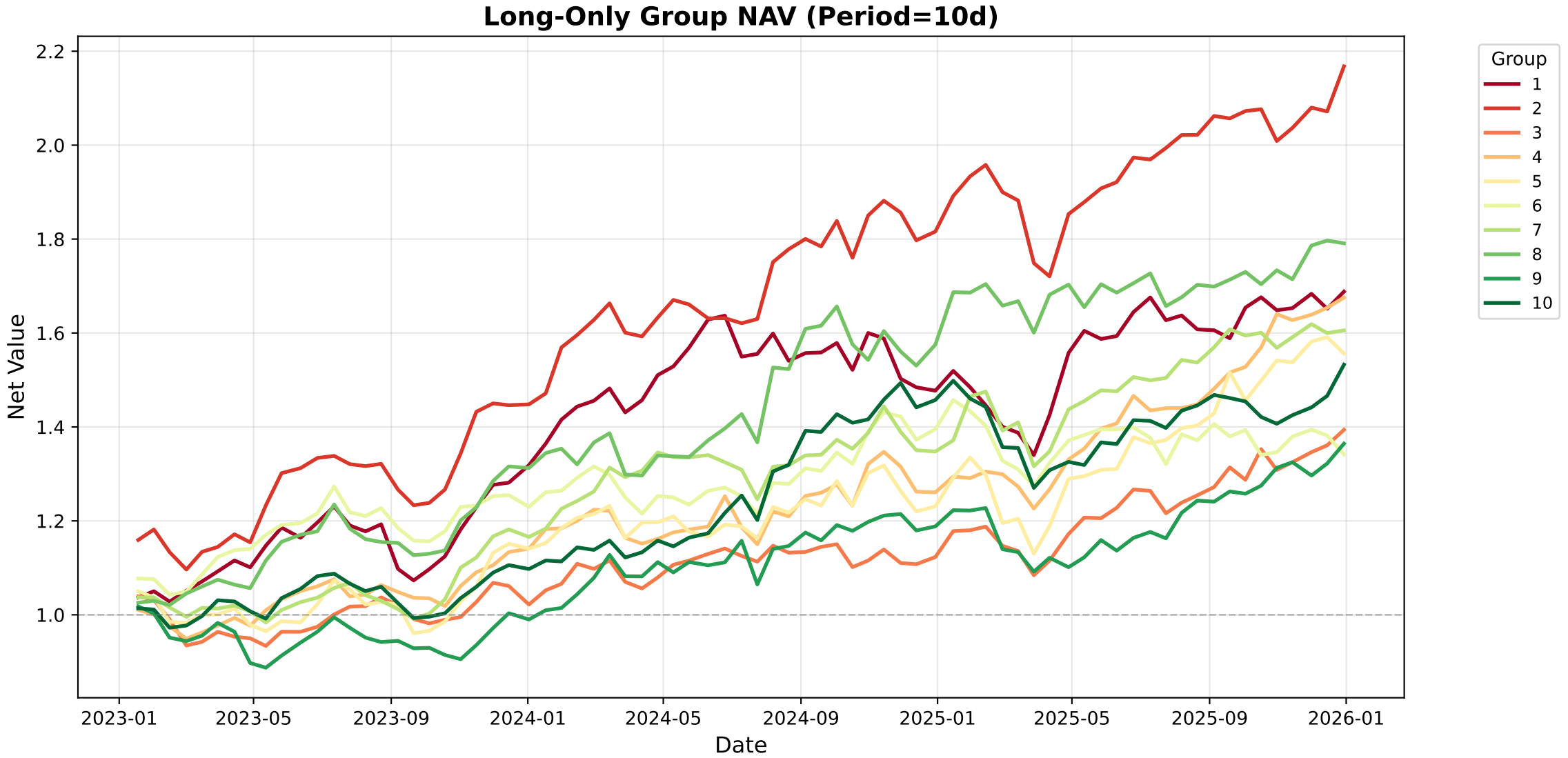

Group Return
(Group 1=lowest factor, Group 10=highest; for momentum Group 10 is typically best)
Group 1
Group 5
Group 10
Return (%)
80
60
40
20
0
-20
1
5
10
Period

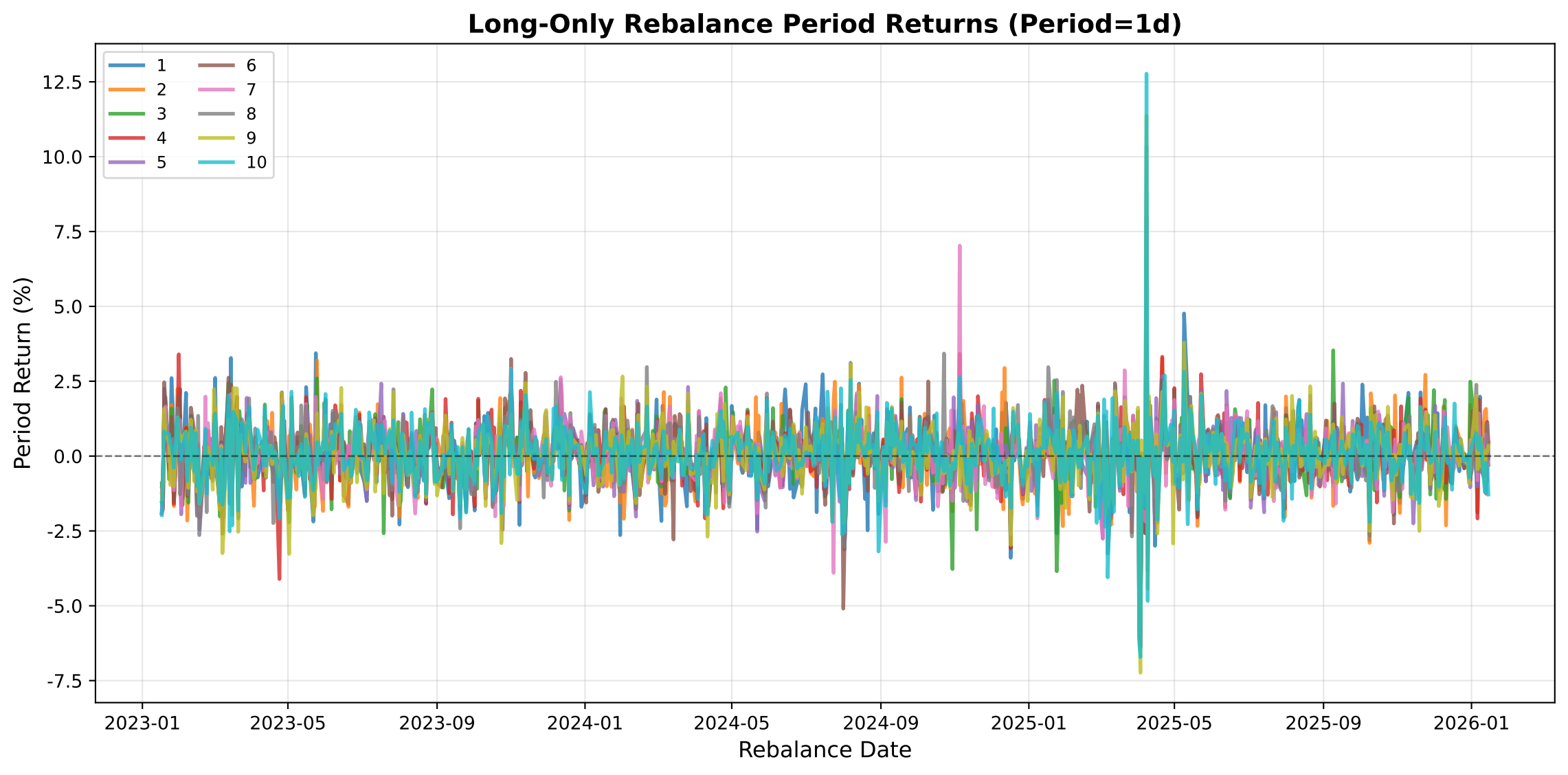
# Long-Only Performance by Group

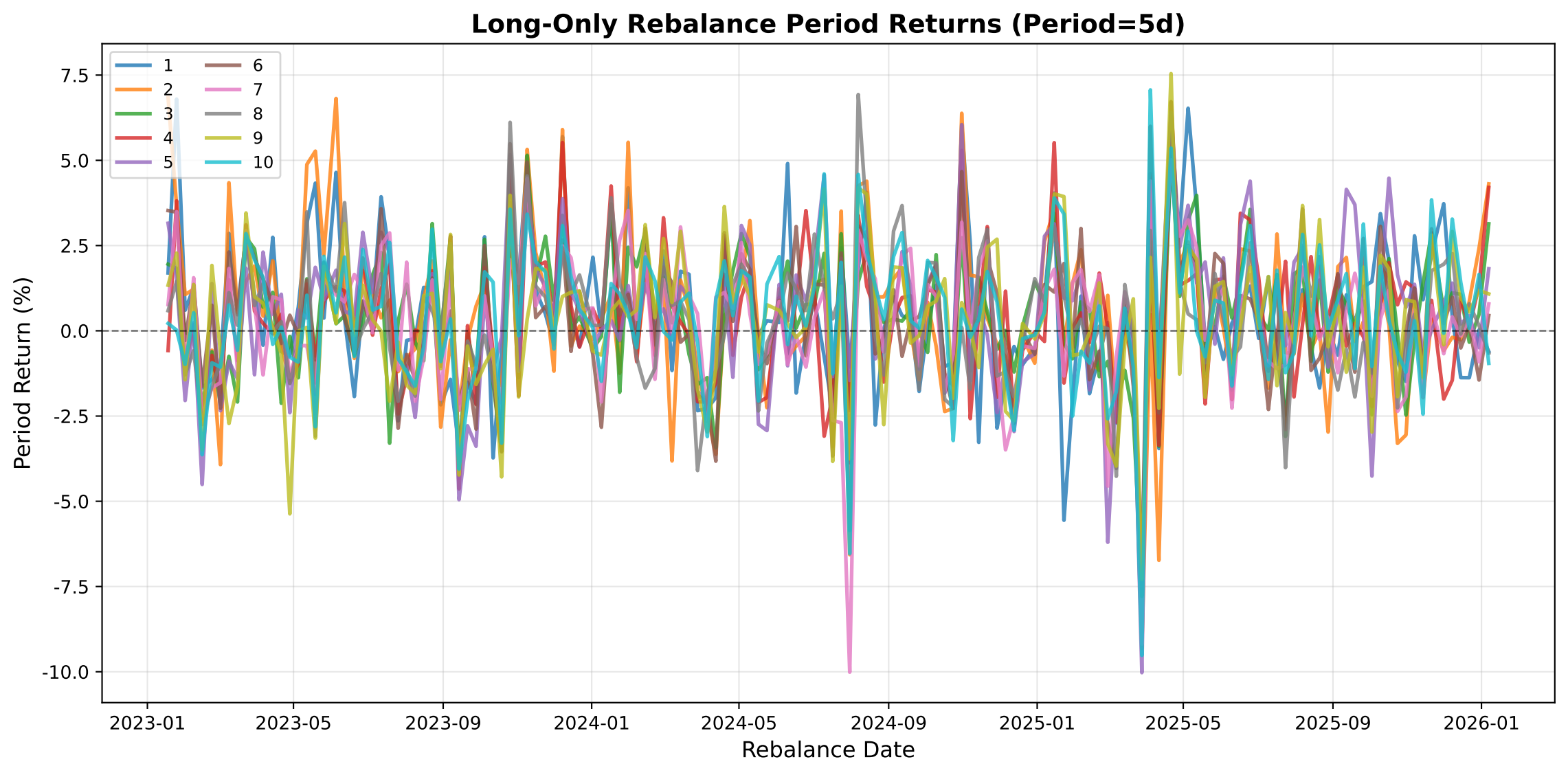
| Period | Group | AReturnRate | AVolatility | SharpeRatio | MaxDrawdown | WinningRatio | PnLRatio |
|---|---|---|---|---|---|---|---|
| Period_1 | G1 | 0.0440 | 0.1582 | 0.2917 | -0.1941 | 0.5073 | 1.0262 |
| Period_1 | G2 | -0.0669 | 0.1543 | -0.5630 | -0.3042 | 0.4927 | 0.9601 |
| Period_1 | G3 | -0.0631 | 0.1487 | -0.5586 | -0.3285 | 0.4807 | 1.0119 |
| Period_1 | G4 | -0.1226 | 0.1457 | -0.9787 | -0.3766 | 0.4834 | 0.9204 |
| Period_1 | G5 | -0.0581 | 0.1530 | -0.5107 | -0.3135 | 0.4993 | 0.9432 |
| Period_1 | G6 | -0.0797 | 0.1566 | -0.6366 | -0.3330 | 0.4954 | 0.9307 |
| Period_1 | G7 | -0.0874 | 0.1532 | -0.7010 | -0.3268 | 0.4900 | 0.9424 |
| Period_1 | G8 | -0.0702 | 0.1613 | -0.5589 | -0.2413 | 0.4821 | 0.9985 |
| Period_1 | G9 | -0.0710 | 0.1716 | -0.5306 | -0.3332 | 0.5086 | 0.9022 |
| Period_1 | G10 | -0.0803 | 0.1708 | -0.5873 | -0.3133 | 0.5126 | 0.8738 |
| Period_10 | G1 | 4.1418 | 0.4718 | 8.7371 | -0.1814 | 0.6400 | 1.0342 |
| Period_10 | G2 | 7.1906 | 0.5330 | 13.4539 | -0.1212 | 0.6667 | 1.2992 |
| Period_10 | G3 | 1.6820 | 0.3892 | 4.2704 | -0.1016 | 0.6133 | 1.0111 |
| Period_10 | G4 | 4.4683 | 0.3963 | 11.2249 | -0.0897 | 0.6667 | 1.0396 |
| Period_10 | G5 | 2.7551 | 0.5178 | 5.2826 | -0.1535 | 0.6267 | 0.9883 |
| Period_10 | G6 | 1.0891 | 0.4246 | 2.5182 | -0.1256 | 0.5600 | 1.1697 |
| Period_10 | G7 | 3.3425 | 0.4053 | 8.1980 | -0.1079 | 0.6133 | 1.2426 |
| Period_10 | G8 | 5.5060 | 0.4697 | 11.6800 | -0.0876 | 0.6000 | 1.4137 |
| Period_10 | G9 | 1.6764 | 0.4439 | 3.7315 | -0.1282 | 0.5867 | 1.0604 |
| Period_10 | G10 | 3.0096 | 0.4070 | 7.3459 | -0.1523 | 0.5867 | 1.2863 |
| Period_5 | G1 | 1.5327 | 0.3631 | 4.1657 | -0.2012 | 0.5467 | 1.3287 |
| Period_5 | G2 | 1.0100 | 0.3936 | 2.5154 | -0.1942 | 0.5533 | 1.1924 |
| Period_5 | G3 | 1.0539 | 0.3042 | 3.3993 | -0.1625 | 0.5867 | 1.0781 |
| Period_5 | G4 | 0.7348 | 0.3247 | 2.2016 | -0.1243 | 0.5533 | 1.0917 |
| Period_5 | G5 | 1.4191 | 0.3605 | 3.8808 | -0.1843 | 0.6067 | 1.0276 |
| Period_5 | G6 | 0.6207 | 0.3303 | 1.8185 | -0.1642 | 0.5467 | 1.1175 |
| Period_5 | G7 | 0.3600 | 0.3264 | 1.0417 | -0.1786 | 0.5800 | 0.8784 |
| Period_5 | G8 | 0.5479 | 0.3299 | 1.6000 | -0.1247 | 0.5467 | 1.0746 |
| Period_5 | G9 | 0.4705 | 0.3362 | 1.3403 | -0.1992 | 0.5667 | 0.9616 |
| Period_5 | G10 | 1.1573 | 0.3321 | 3.4243 | -0.1741 | 0.5733 | 1.1390 |

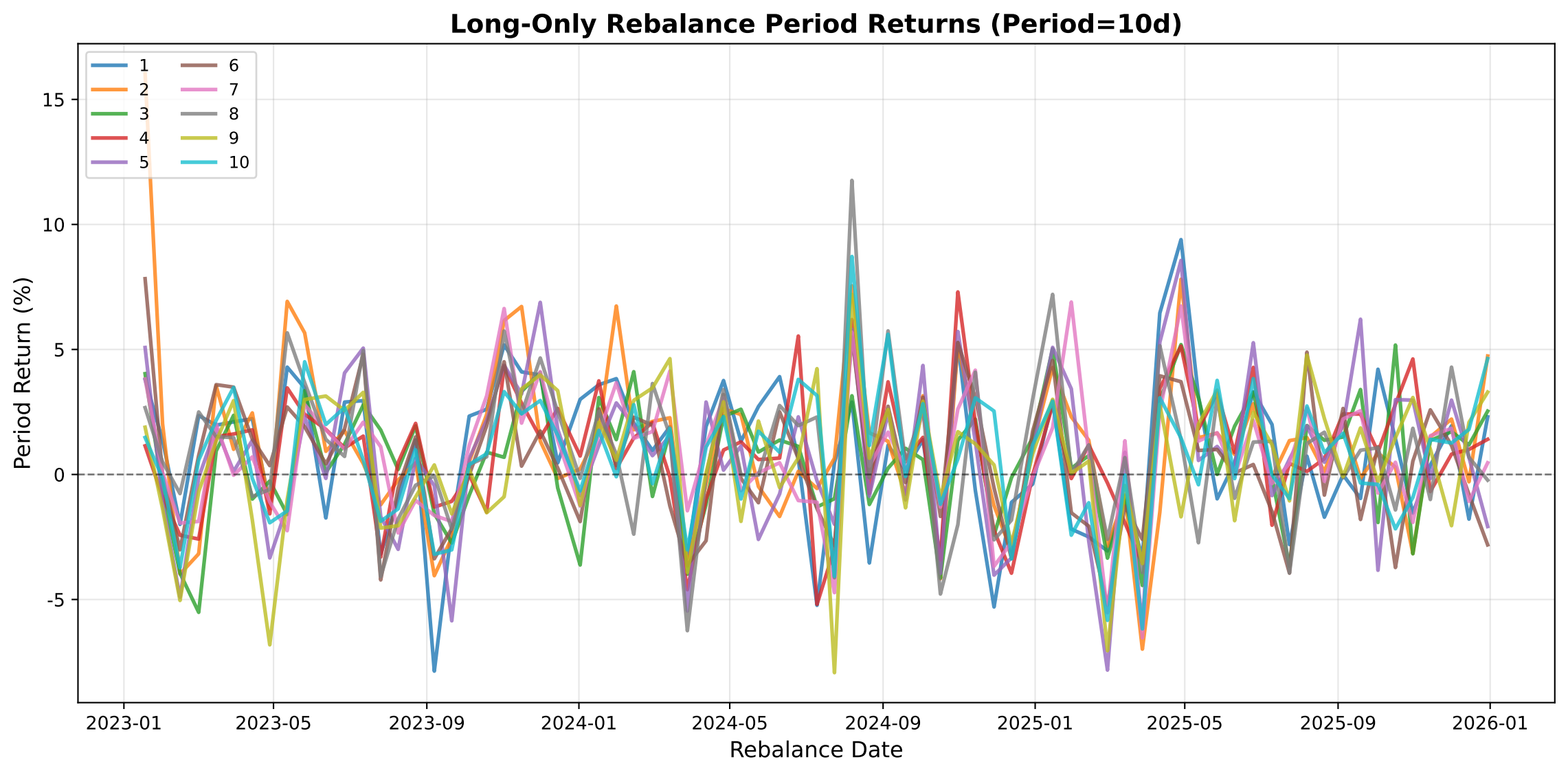

Long-Only Group NAV (Period=1d)
Net Value
1.1
1.0
0.9
0.8
0.7
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
Date
Group
1
2
3
4
5
6
7
8
9
10


Long-Only Group NAV (Period=5d)
Net Value
Date
1.8
1.6
1.4
1.2
1.0
0.8
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
Group
1
2
3
4
5
6
7
8
9
10


Long-Only Group NAV (Period=10d)
Net Value
Date
2.2
2.0
1.8
1.6
1.4
1.2
1.0
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
Group
1
2
3
4
5
6
7
8
9
10


Long-Only Rebalance Period Returns (Period=1d)
1
2
3
4
5
6
7
8
9
10
12.5
10.0
7.5
5.0
2.5
0.0
-2.5
-5.0
-7.5
Period Return (%)
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
Rebalance Date


Long-Only Rebalance Period Returns (Period=5d)
1
2
3
4
5
6
7
8
9
10
7.5
5.0
2.5
0.0
-2.5
-5.0
-7.5
-10.0
Period Return (%)
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
Rebalance Date


Long-Only Rebalance Period Returns (Period=10d)
1
2
3
4
5
6
7
8
9
10
15
10
5
0
-5
Period Return (%)
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
Rebalance Date